NYU|Tandon

MA 2034 WORKSHEET 1 FALL 2026

Differential Equations Worksheet: Exponential Models

Total: 100 points

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Instructions: Show all work and explain your reasoning when requested. **You must upload your work on this template and assign your work to the appropriate questions.**

Question 1. Alligator Population — Exponential Growth (40 points)

A protected population of alligators is modeled by

$$\frac{dP}{dt} = kP,$$

where $P(t)$ is the number of alligators t years after conservation efforts begin. Initially there are 400 alligators. After 5 years, the population has increased to 500.

(a) Interpretation of the Model (5 points)

Cirlce ALL true statements that apply to the differential equation

$$\frac{dP}{dt} = kP.$$

A. The alligator population increases by exactly k alligators each year.

B. The rate at which the alligator population changes is proportional to the current population.

C. The alligator population will eventually approach k alligators.

D. The percentage growth of the population increases at a constant rate.

E. The alligator population increases only when $P = k$.

Choice: B, D

Briefly explain your choice(s).

(b) Solve the Differential Equation (10 points)

Starting with

$$\frac{dP}{dt} = kP,$$

solve the differential equation by separation of variables. Then use the initial condition

$$P(0) = 400$$

to obtain a solution containing only the unknown parameter k.

Show each major step.

$$\frac{dP}{dt} = kP$$

$$P^{-1}\frac{dP}{dt} = k$$

$$\frac{1}{P}\,dP = k\,dt$$

$$\int \frac{1}{P}\,dP = \int k\,dt$$

$$\ln|P| = kt + C$$

$$|P| = e^{kt}e^{C}$$

$$e^{C} = C$$

$$\therefore\ P = Ce^{kt}$$

$$P(0) = 400$$

$$400 = Ce^{k(0)}$$

$$400 = C$$

$$P(t) = 400e^{kt}$$

Question 1 continued (40 points total)

(c) Determine the Growth Constant (5 points)

Use the additional information

$$P(5) = 500$$

to determine k. Write the complete particular solution $P(t)$. Show all your work.

$500 = 400e^{5k}$

$\frac{5}{4} = e^{5k}$

$\ln\left(\frac{5}{4}\right) = 5k$

$k = \frac{\ln(1.25)}{5}$

(d) Predict the Population (5 points)

According to this model, approximately how many alligators will there be after 15 years? Show all your work.

$P(15) = 400e^{(15)\frac{\ln(\frac{5}{4})}{5}}$

$P(15) = 400\left(\frac{5}{4}\right)^3$

$\Rightarrow \frac{400}{64} \times 125 = 781.25$

$\therefore \approx 781$ alligators in 15 years

(e) Changing the Parameter k (10 points)

Suppose improved habitat causes the value of k to **double**, while the initial population remains 400.

1. Describe how doubling k affects the graph of $P(t)$.

$P(t) = 400e^{2kt}$ starts same but rises e^{kt} faster
∴ it is more steep, faster for more population

2. Would doubling k cause the population to be twice as large at every future time? Explain mathematically or conceptually.

No because graph is rising e^{kt} fastll not 2x faster
∴ not fast enough

(f) Long-Term Behavior (5 points)

Determine

$$\lim_{t\to\infty} P(t).$$

Explain why this long-term prediction may be unrealistic for an actual alligator population.

Because there isn't infinite food/space and eventually once population is back, efforts stop

Question 2. A Message from Space — Exponential Decay (30 points)

A distant spacecraft sends a continuous radio signal toward Earth. As its power source deteriorates, the strength $S(t)$ of the transmitted signal is modeled by

$$\frac{dS}{dt} = -kS, \qquad k > 0.$$

Initially the signal strength is 80 units. Six years later it has fallen to 50 units.

(a) Find the Model (10 points)

Find k and determine the particular solution $S(t)$. Show enough work to justify your answer.

$\frac{dS}{dt} = -kS$

$\frac{dS}{S} = -k\,dt$

$\int \frac{1}{S}\,dS = \int -k\,dt$

$\ln|S| = -kt + C$

$S = Ce^{-kt}$

$S(0) = 80 = Ce^{-k(0)}$

$\therefore 80 = C$

$C = 80$

$S(6) = 50$

$50 = 80e^{-k(6)}$

$k = -\frac{1}{6}\ln\left(\frac{5}{8}\right)$

(b) Signal Strength of 10 Units (5 points)

According to the model, when will the signal strength fall to 10 units? Show all your work.

$10 = 80e^{-kt}$

$\frac{1}{8} = e^{-kt}$

$+\ln(8) = +\frac{\ln\left(\frac{8}{5}\right)}{6}t$

$t = \frac{6\ln(8)}{\ln\left(\frac{8}{5}\right)}$

$= 26.5$ years

$\therefore \approx 26.5$

(c) Changing the Decay Constant (10 points)

Engineers redesign the power system so that the decay constant k is reduced by 50%.

1. Describe how $S(t)$ changes.

$0.5k \therefore S(t) = 80e^{-(\frac{k}{2})t}$

$\therefore$ it is slower and is above original curve

2. What happens to the time required for the signal to fall to one-half of its initial strength? Explain.

$40 = 80e^{-kt} \Rightarrow 0.5e^{-kt}$

$\therefore -\ln(0.5) = -kt$

$t = \frac{\ln(2)}{k}$

$\Rightarrow t = \frac{\ln(2)}{k \times \frac{1}{2}} \Rightarrow \frac{2\ln 2}{k} = 2t$

$\therefore$ double

(d) Long-Term Behavior (5 points)

Determine

$$\lim_{t \to \infty} S(t).$$

Does the mathematical model predict that the signal ever becomes exactly zero at a finite time? Explain.

$e^{-kt} \to 0$ as $t \to \infty$

$\therefore \lim_{t \to \infty} S(t) = 0$

$S(t) \neq 0$ since $80e^{-kt} > 0$

Question 3. A Meteorite in the Laboratory — Newton's Law of Heating & Cooling (30 points)

A meteorite recovered from Antarctica has been stored outside at $-20°\text{C}$. It is brought into a laboratory maintained at $20°\text{C}$.

Its temperature is modeled by Newton's law of heating:

$$\frac{dT}{dt} = k(20 - T), \qquad k > 0.$$

When the meteorite enters the laboratory,

$$T(0) = -20°\text{C}.$$

After 30 minutes,

$$T(30) = 0°\text{C}.$$

(a) Find the Model (10 points)

Determine k and find the particular solution $T(t)$. Show all your work.

$\frac{dT}{dt} = k(20-T)$

$\int \frac{1}{20-T}\,dT = \int k\,dt$

$\Rightarrow -\ln|20-T| = kt + C$

$20 - T = Ce^{-kt}$

$T = 20 - Ce^{-kt}$

$-20 = 20 - Ce^{0}$

$\therefore C = -40$

$0 = 20 - 40e^{-kt}$

$\Rightarrow \ln\left(\frac{1}{2}\right) = -30k$

$-\ln(2) = -30k$

$k = \frac{\ln(2)}{30}$

$T(t) = 20 - 40e^{-\frac{\ln 2}{30}t}$

(b) Reach 15°C (5 points)

According to the model, how long will it take the meteorite to reach 15°C? Show all your work.

$T(t) = 15$

$\therefore 15 = 20 - 40e^{-kt}$

$\Rightarrow -5 = -40e^{-kt}$

$\Rightarrow \frac{1}{8} = e^{-kt}$

$-\ln(8) = -kt$

$\ln(8) = kt$

$t = \frac{\ln 8}{k}$

$t = \frac{30\ln(8)}{\ln 2}$

$t = 90$

(c) Increasing k (5 points)

Suppose a fan is pointed at the meteorite, causing k to increase while the laboratory remains at 20°C.

Describe how increasing k affects the graph of $T(t)$.

Does increasing k change the eventual temperature of the meteorite? Explain.

larger k makes e^{-kt} decrease faster
∴ it gets a steeper slope but
since at $\lim_{t\to\infty} T(t) = 20°C$ (eventual temp does not change)

(d) Changing the Room Temperature (4 points)

Now suppose the laboratory temperature is increased from 20°C to 25°C.

Without solving the new differential equation, explain what happens to the long-term behavior of the solution and determine

$$\lim_{t\to\infty} T(t).$$

at 25°C it becomes $\frac{dT}{dt} = k(25 - T)$

∴ eventual temp = 25°C

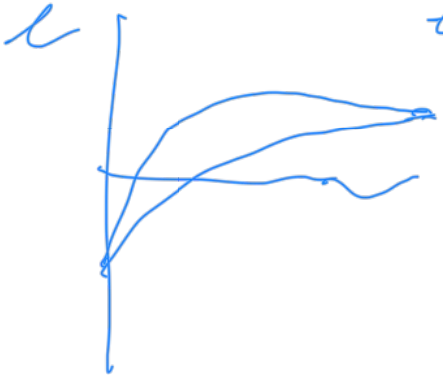
(e) Sketch Solutions (6 points)

Include a sketch of a possible solution, $T(t)$, for each of the curves in parts (a), (c) and (d).

a



c



d

